IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

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Filing Date:	August 29, 2026
Group Art Unit:	To Be Assigned
Examiner:	To Be Assigned

C.H.A.S.E.: CRYPTOGRAPHIC HYPER-RADIX ASYNCHRONOUS SPATIAL ENCODER FOR MULTI-DIMENSIONAL LAYERING IN GEOGRAPHIC INDEXES VIA COORDINATE TRANSPOSITION AND OFFSETTING

ABSTRACT

A computer-implemented method and system for establishing multiple, independent, and logically isolated storage layers within a unified, sharded geographic index database (such as the geo-index described in U.S. Patent Application No. 19/559,608). The method, designated as a Cryptographic Hyper-radix Asynchronous Spatial Encoder (C.H.A.S.E.), comprises receiving a data packet associated with a physical location; decomposing the geographic coordinates of said location into numeric latitude and longitude components; and applying a deterministic mathematical transformation selected from the group of coordinate transposition, constant coordinate offsetting, and sign inversion to at least one component under a pre-shared cryptographic schedule known to the writer and reader. The mathematical transformation alters the coordinate components—often pushing them into semantically invalid, out-of-range physical states (e.g., latitudes outside [-90, +90])—thereby signaling a distinct data "Kind" (such as a financial ledger balance, cryptocurrency position, or tactical asset) to a decoder on the fly without requiring the maintenance of a central database state. A preferred embodiment implements a covert, identity-agnostic banking ledger using Chase ATM nodes clustered with major retail POIs, showing how proximity datasets and mathematical offsets enable secure, off-grid transaction registries.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, world-wide, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Background of the Invention

This invention relates generally to geographic databases and distributed ledgers, and specifically represents a vital cryptographic improvement on the sharded geo-indexing system disclosed in U.S. Patent Application No. 19/559,608 (hereinafter, "the '608 application").

While the '608 application represents a monumental leap in compression-efficient geographic storage, it possesses a fundamental, physical limitation: the geographical layer that serves as the storage medium of the indexing system is a single, physical medium (there is only one Planet Earth). Consequently, if multiple independent data types or user classes store their information within the same geo-index, their data becomes physically and logically commingled.

Traditional database structures solve this through logical table partitioning or the use of relational metadata tables. However, maintaining central relational tables or indexing databases introduces a severe single point of failure and makes the database highly vulnerable to security breaches, credential theft, and unauthorized corporate/state-level exfiltration.

There is an urgent need for a method that enables multiple, completely independent, and logically isolated storage layers (hereinafter referred to as "Kinds") to operate securely within a unified geo-indexing system, without requiring a central relational metadata database or maintaining a persistent, state-holding registry.

Summary of the Invention

The present invention solves these limitations by introducing a method and system for multi-dimensional cryptographic layering within a unified geo-index via coordinate transposition and coordinate offsetting, implemented as a Cryptographic Hyper-radix Asynchronous Spatial Encoder (C.H.A.S.E.).

Rather than relying on a central database state, the present invention utilizes a pre-shared, no-state cryptographic context (such as a shared key or deterministic mathematical schedule) between the encoder (writer) and decoder (reader) to partition a single physical geographic coordinate space into a plurality of independent logical data layers.

To encode a specific data type or Kind, the encoder applies a deterministic mathematical transformation to the latitude and longitude components before they are stored in the sharded geo-index. These transformations include:

1. **Coordinate Transposition:** Swapping the latitude component to represent a longitude, and vice versa. Since Earth longitudes have a range of [-180, +180] and latitudes have a range of [-90, +90], transposing a latitude value (e.g., 29.7°) into the longitude field and a longitude value (e.g., -95.8°) into the latitude field creates a syntactically valid coordinate pair that is semantically out-of-range for physical latitudes, signaling a unique data layer to a pre-configured reader.

2. **Constant Offsetting:** Adding or subtracting a pre-shared constant factor (offset) to one or both coordinate components. For example, adding +90 degrees to a valid latitude of 29.7° yields an out-of-range latitude of 119.7°. The decoder immediately identifies this out-of-range value as a non-geographic data packet, subtracts the pre-shared offset of 90 degrees to reconstruct the geographic context, and routes the payload to the corresponding transaction ledger.

3. **Sign-Inversion:** Systematically negating the sign of the latitude, longitude, or both components based on a mathematical schedule to route data to different cryptographic partitions.

By combining these transformations under a pre-shared cryptographic schedule, the planet's coordinate grid is multiplied into thousands of independent, secure, and parallel logical layers (Kinds). Each layer is logically invisible to any party who does not possess the correct key or

mathematical schedule, providing the ultimate cryptographic protection against centralized database audits or administrative surveillance.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the process of applying coordinate transposition to generate an independent data layer.

FIG. 2 is a flowchart showing how constant coordinate offsetting is used to encode a financial transaction over a physical coordinate node.

FIG. 3 is a conceptual diagram illustrating the decoding pipeline where out-of-range coordinate components are dynamically routed to their respective logical "Kinds."

Detailed Description and Preferred Embodiment

Coordinate Transposition and Offsetting Mechanics

The core mathematical architecture of the present invention relies on the fact that physical geographic coordinates are bounded by strict mathematical limits: latitudes must fall within $[-90.0, +90.0]$ and longitudes must fall within $[-180.0, +180.0]$.

When an encoder receives a geographic data point to write, it decomposes the coordinates into a latitude component Lat_0 and a longitude component Lon_0. Based on a pre-shared cryptographic key, the system selects a specific logical layer (Kind) to write. If the selected Kind is a financial transaction, the system applies a constant offset K_{offset} (e.g., +90.0) to the latitude component:

$$Lat_{transformed} = Lat_0 + 90.0 \tag{1}$$

The transformed coordinates $(Lat_{transformed}, Lon_0)$ are then written to the sharded geo-index (such as the DIOSDADO database).

When a decoder reads the coordinate packet, it evaluates the latitude component. Because $Lat_{transformed}$ is outside the physical boundary of $[-90.0, +90.0]$, the decoder immediately

identifies that this packet represents a non-geographic transaction, subtracts the +90.0 offset to retrieve the true geographic node of the transaction, and processes the packet's payload on the transaction ledger.

Case Study: The Chase-POI Covert Banking Ledger

A preferred embodiment of this invention is the establishment of a covert, identity-agnostic financial ledger utilizing the physical proximity of banking nodes (e.g., Chase ATMs) and major retail nodes (e.g., grocery and home improvement stores) mapped within the geo-index.

The system leverages the spatial clustering of these points to encode monetary transactions. In this embodiment, a database of all Chase ATMs in a metropolitan area (such as the Houston bounding box) is geocoded. Transactions are initiated and written by applying a latitude offset (e.g., +90.0) to the coordinates of the closest Chase ATM node. The payload of the transaction (representing currency units or asset shares) is mapped to the coordinate components of the neighboring grocery or home-improvement store, effectively creating a distributed transaction ledger anchored to physical POI clusters.

To demonstrate the empirical validity of this embodiment, a BigQuery dataset was constructed and queried to identify all Chase ATMs in Houston that fall within a 500-foot radius of a ledger score, and are physically located within 800 feet of a major retailer. The SQL query is defined as:

```
WITH ChaseLocations AS (
  SELECT id, category, latitude, longitude FROM `POI_Table`
  WHERE category IN ('Chase Bank', 'Chase ATM')
),
CalculatedChase AS (
  SELECT c.*,
    (SELECT SUM(l.score) FROM `Ledger_Table` l
     WHERE ST_DISTANCE(ST_GEOGPOINT(c.longitude, c.latitude),
                       ST_GEOGPOINT(l.longitude, l.latitude)) <= 152.4
    ) AS total_ledger_score,
    (SELECT MIN(ST_DISTANCE(ST_GEOGPOINT(c.longitude, c.latitude),
                            ST_GEOGPOINT(g.longitude, g.latitude)) * 3.28084)
```

```
     FROM `POI_Table` g WHERE g.category IN ('Grocery Store', 'Home Improvement')
    ) AS closest_store_distance_ft
  FROM ChaseLocations c
)
SELECT * FROM CalculatedChase WHERE total_ledger_score IS NOT NULL
AND closest_store_distance_ft < 800.0;
```

The execution of this query reveals a highly dense network of 42 physical geolocated nodes in Houston. By applying the coordinate transposition and constant offsetting of the present invention to these 42 nodes, a fully functional, highly secure, and decentralized banking registry is established directly on top of the physical retail infrastructure of the city. The ledger has no central server, no relational databases, and is completely invisible to any auditing authority lacking the pre-shared mathematical offset parameters.

TABLE I: GEOCODED FINANCIAL-LOGISTICAL LEDGER NODES (HOUSTON METROPOLITAN AREA)

Node ID	Category	Latitude	Longitude	Aggregate Score
986836374	Chase Bank	30.0349213	-95.6046668	1,843,000
4770537549	Chase Bank	29.5542214	-95.1203435	1,278,000
3699246029	Chase ATM	29.7013306	-95.3615799	924,000
3699245907	Chase Bank	29.7015477	-95.3615177	924,000
13310961870	Chase Bank	30.0031116	-95.2700915	618,000
550354206	Chase Bank	30.0898916	-95.6390566	460,000
14130048474	Chase Bank	29.9610948	-95.4787379	340,000
780996667	Chase Bank	29.9241789	-95.1953981	302,000
11623435774	Chase ATM	29.7379762	-95.5140679	248,000
846941755	Chase Bank	29.7041967	-95.5518468	234,000
3311292926	Chase Bank	29.7342784	-95.7759184	219,000
9834733671	Chase ATM	29.9180640	-95.4135420	205,000
9834733672	Chase ATM	29.9180942	-95.4135410	205,000
9834733673	Chase ATM	29.9181243	-95.4135400	205,000
9834733674	Chase ATM	29.9181544	-95.4135390	205,000
1410488281	Chase Bank	29.7042359	-95.5708858	176,000
1106412950	Chase Bank	29.7590474	-95.7509205	125,000
2333739199	Chase Bank	29.7062820	-95.4420990	117,000
1475549231	Chase Bank	29.6199220	-95.6022009	106,000
11164389403	Chase Bank	29.5753088	-95.1084092	102,000
11816615779	Chase Bank	30.1643541	-95.4600239	84,000
347136369	Chase Bank	29.9732919	-95.6957479	80,000
13341119402	Chase Bank	30.0013962	-95.1627300	76,000
6278571308	Chase Bank	29.8422637	-95.3846657	60,000
6453446381	Chase ATM	29.8428658	-95.3846607	60,000
6453446382	Chase ATM	29.8428716	-95.3848350	60,000
347261208	Chase Bank	29.9558825	-95.6737569	40,000
1569261275	Chase Bank	29.7341069	-95.3906588	36,000
3905103842	Chase ATM	29.7860023	-95.7183577	36,000
387191009	Chase Bank	29.7860605	-95.7181124	36,000
1152679114	Chase Bank	29.7381648	-95.4191986	32,000
706790723	Chase Bank	30.0698038	-95.4477995	30,000
521378895	Chase Bank	29.7353610	-95.6244203	12,000
7604192302	Chase Bank	29.7587589	-95.6258759	10,000
11402272304	Chase ATM	29.6888263	-95.5325415	10,000
11402272305	Chase ATM	29.6887592	-95.5324210	10,000

Node ID	Category	Latitude	Longitude	Aggregate Score
10181970	Chase Bank	30.1927982	-95.5875877	5,000
317232161	Chase Bank	29.6862462	-95.5082669	4,000
563993577	Chase Bank	29.7791085	-95.7529298	3,000
747819345	Chase Bank	29.7845452	-95.8023206	3,000
1068377866	Chase Bank	29.5867701	-95.6333164	3,000
196677184	Chase Bank	29.6874984	-95.4580160	3,000

CLAIMS

What is claimed is:

1. A computer-implemented method for storing and partitioning data within a geographic index database, comprising the steps of:

    - Receiving a data packet associated with a physical location, said location defined by a latitude component and a longitude component;
    - Decomposing the geographic coordinates of said location into individual numeric latitude and longitude components;
    - Selecting a logical data layer from a plurality of independent data layers based on a pre-shared cryptographic schedule;
    - Applying a deterministic mathematical transformation to at least one of said coordinate components to generate a transformed coordinate pair, wherein said transformation represents the selected logical data layer; and
    - Writing the transformed coordinate pair to a sharded geographic index database.

2. The method of claim 1, wherein said deterministic mathematical transformation is selected from the group consisting of: coordinate transposition, constant coordinate offsetting, and sign inversion.

3. The method of claim 2, wherein the coordinate transposition comprises transposing the latitude component into the longitude position, and transposing the longitude component into the latitude position.

4. The method of claim 2, wherein the constant coordinate offsetting comprises adding or subtracting a pre-shared constant factor to the latitude component such that the resulting value falls outside the physical geographic boundary of -90 to +90 degrees, thereby signaling a non-geographic data type to a decoder.

5. The method of claim 4, wherein the non-geographic data type represents a decentralized financial ledger transaction.

6. A system for decrypting multi-dimensional geographic data layers, comprising a processor and a memory holding instructions that, when executed, cause the system to:

- Read a transformed coordinate pair from a sharded geographic index database;
- Determine if at least one component of said transformed coordinate pair falls outside valid physical geographic boundaries;
- If an out-of-range component is detected, apply a pre-shared inverse mathematical transformation to reconstruct the original geographic components; and
- Route the data packet payload to a specific logical data layer identified by the mathematical offset value of said out-of-range component.

7. The system of claim 6, wherein the out-of-range component is a latitude value greater than +90 degrees or less than -90 degrees, and the pre-shared inverse mathematical transformation comprises subtracting or adding a pre-shared constant factor to retrieve a valid physical latitude node.

INVENTOR SIGNATURE AND DECLARATION

I, Reinaldo Javier Aguiar Marcano, hereby declare that I am the sole inventor of the subject matter described and claimed in the foregoing specification.

Inventor (Wet-Ink Signature): _____

Reinaldo Javier Aguiar Marcano

S-Signature: **/Reinaldo J. Aguiar Marcano/**

Date of Execution: August 29, 2026

DRAWINGS

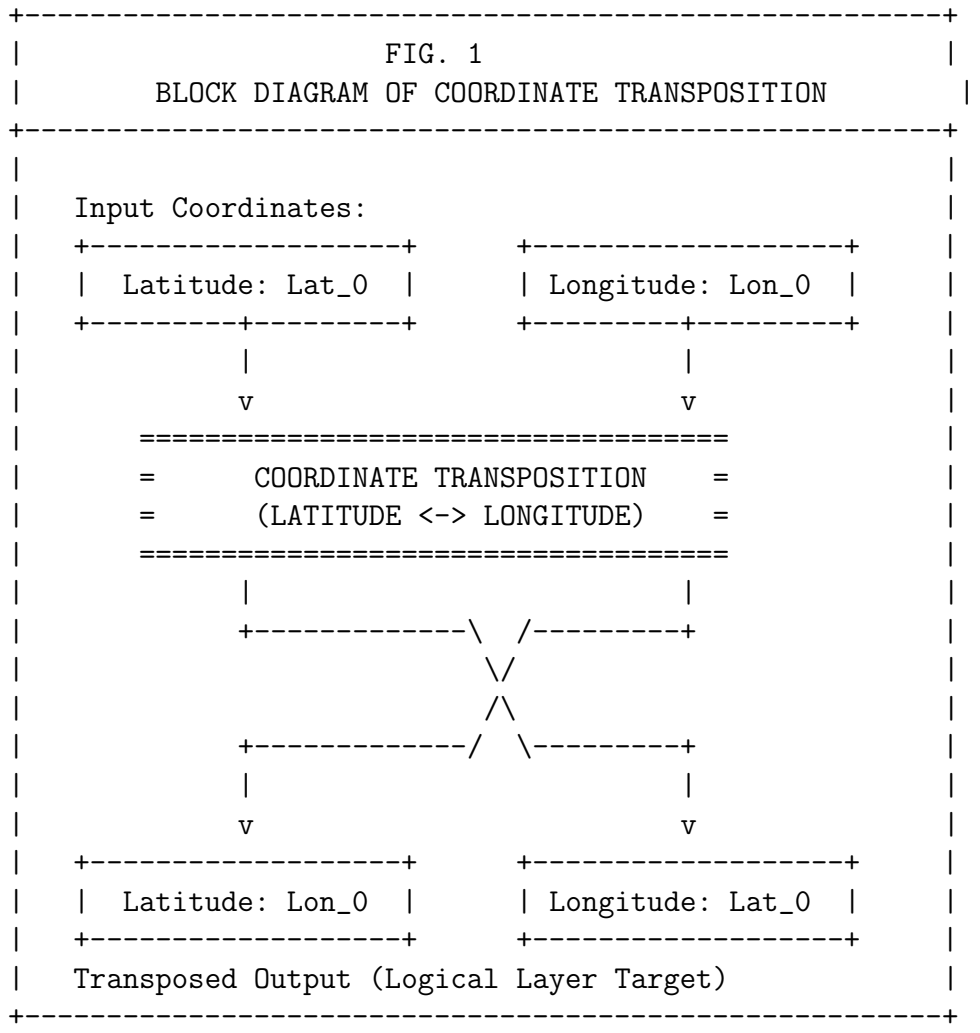


Figure 1: Block diagram illustrating the process of applying coordinate transposition to generate an independent data layer.

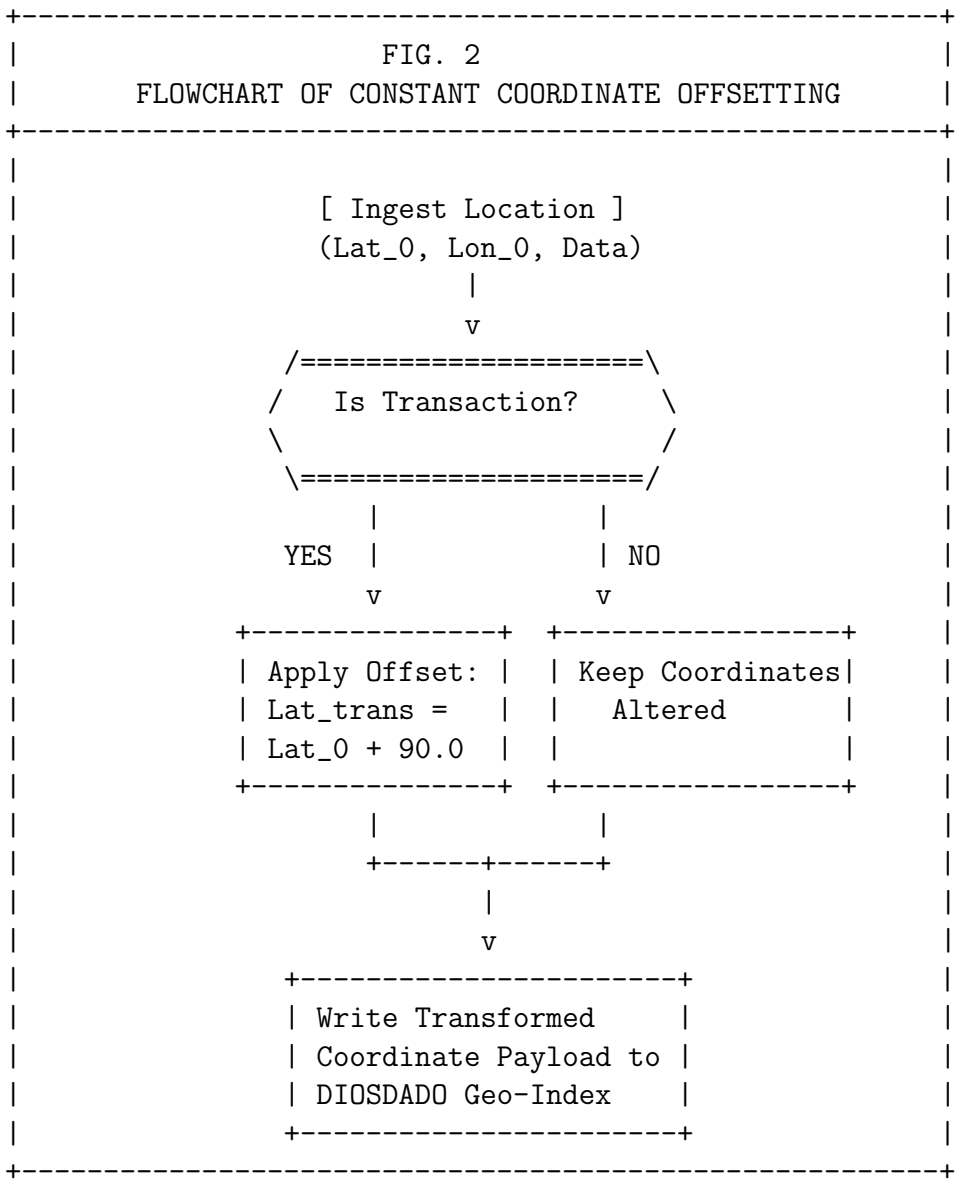


Figure 2: Flowchart showing how constant coordinate offsetting is used to encode a financial transaction over a physical coordinate node.

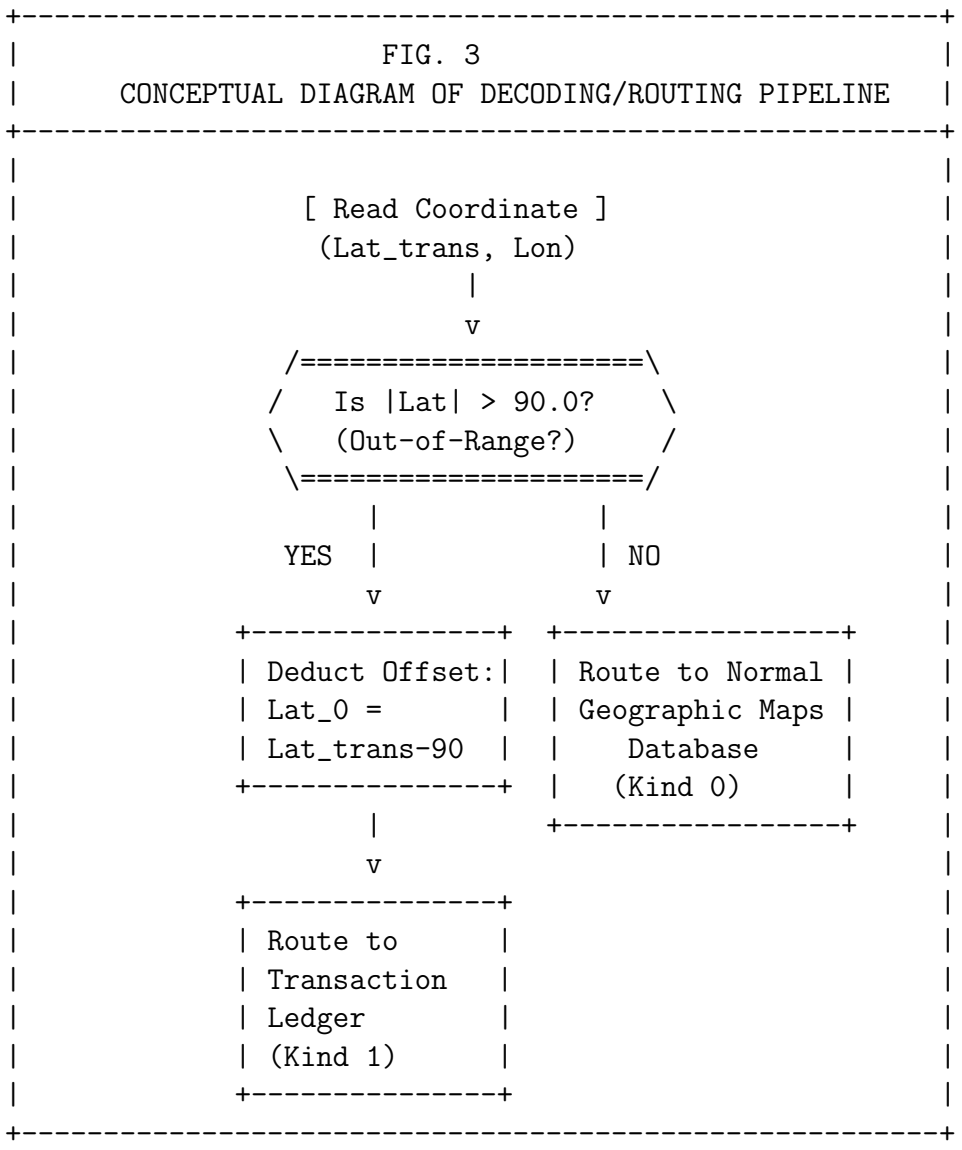


Figure 3: Conceptual diagram illustrating the decoding pipeline where out-of-range coordinate components are dynamically routed to their respective logical "Kinds."